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A high-resolution PheWAS approach to alcohol-related polygenic risk scores reveals mechanistic influences of alcohol reinforcing value and drinking motives

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Abstract

Aims: This study uses a high-resolution phenome-wide approach to evaluate the motivational mechanisms of polygenic risk scores (PRSs) that have been robustly associated with coarse alcohol phenotypes in large-scale studies.

Methods: In a community-based sample of 1534 Europeans, we examined genome-wide PRSs for the Alcohol Use Disorders Identification Test (AUDIT), drinks per week, alcohol use disorder (AUD), problematic alcohol use (PAU), and general addiction, in relation to 42 curated phenotypes. The curated phenotypes were in seven categories: alcohol consumption, alcohol reinforcing value, drinking motives, other addictive behaviors, commonly comorbid psychiatric syndromes, impulsivity, and personality traits.

Results: The PRS for each alcohol phenotype was validated via its within-sample association with the corresponding phenotype (adjusted R^2s = 0.35–1.68%, *P*s = 0.012–3.6 × 10^{-7}) with the exception of AUD. All PRSs were positively associated with alcohol reinforcing value and drinking motives, with the strongest effects from AUDIT-consumption (adjusted R^2s = 0.45–1.33%, *P*s = 0.006–3.6 × 10^{-5}) and drinks per week PRSs (adjusted R^2s = 0.52–2.28%, *P*s = 0.004–6.6 × 10^{-9}). Furthermore, the PAU and drinks per week PRSs were positively associated with adverse childhood experiences (adjusted R^2s = 0.6–0.7%, *P*s = 0.0001–4.8 × 10^{-4}).

Conclusions: These results implicate alcohol reinforcing value and drinking motives as genetically-influenced mechanisms using PRSs for the first time. The findings also highlight the value of dissecting genetic influence on alcohol involvement through diverse phenotypic risk pathways but also the need for future studies with both phenotypic richness and larger samples.

Keywords: alcohol use; alcohol use disorder identification test (AUDIT); genetics; genomics; phenome-wide association study (PheWAS); problematic alcohol use; polygenic risk score (PRS)

Introduction

The influence of genetic factors on alcohol use and alcohol use disorder (AUD) has long been recognized in twin studies (Tsuang et al. 2001). However, robust results from large-scale genome-wide association studies (GWASs) on alcohol phenotypes have only become available recently. These large genetic studies are a first step toward pinpointing the most plausible genetic determinants of alcohol use (Clarke et al. 2017; Kranzler et al. 2019; Liu et al. 2019; Saunders et al. 2022). In particular, there have been considerable efforts to conduct large-scale GWASs of problematic alcohol use (PAU; Gelernter et al. 2014; Walters et al. 2018; Zhou et al. 2020), either as a binary categorization or as the total score and sub-component scores from the Alcohol Use Disorders Identification Test (AUDIT) (Sanchez-Roige et al. 2019a, 2019c; Mallard et al. 2022), which is a screening measure of PAU. Intriguingly, the two sub-scales of AUDIT, the 3-item consumption index (AUDIT-C) and the 7-item problematic use index (AUDIT-P), exhibit distinctive patterns of association with genetic determinants, both in terms of the strongest association signals and their overall genetic correlation with other related phenotypes (Sanchez-Roige et al. 2019a, 2019c). This along with other studies (Kranzler et al. 2019; Zhou et al. 2020) suggests that alcohol consumption and problematic use are not genetically identical constructs.

The next step toward improving our understanding of genetic signals associated with alcohol use can be through identifying their shared or unique associations with motivational traits that are either directly or peripherally related to alcohol use and AUD. Phenome-wide association studies (PheWASs), whereby variants, genes, or PRSs of interest are tested across a large number of different phenotypes, can be a useful strategy to parse GWAS findings derived from coarser phenotypes. This entails testing established genetic signals associated with alcohol use, either individually or collectively as an aggregate genetic score, in relation to high-resolution phenotypes from phenotypic domains that have been linked to alcohol consumption and problematic use in existing observational studies. The contributing mechanisms to alcohol use are multifaceted and can improve our understanding both the appeal of alcohol and the potential for misuse or addiction (MacKillop et al. 2022).

There are several well-established alcohol-specific mechanisms, including alcohol involvement, alcohol reinforcing

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value (Murphy and MacKillop 2006) (i.e. the motivational significance of alcohol as a reinforcer), and drinking motives (Céspedes et al. 2021) (i.e. self-reported reasons for drinking). Furthermore, alcohol use is often comorbid with other conditions, including concurrent addictive behavior (Cavicchioli et al. 2018) (i.e. other substance use and compulsive eating); and common psychiatric syndromes (Ohlmeier et al. 2008; Blanco et al. 2013; Debell et al. 2014; Kathryn Mchugh and Weiss 2019) (i.e. depression, posttraumatic stress disorder [PTSD], and attention-deficit/ hyperactivity disorder [ADHD]) and adverse childhood experience (ACE) (Dube et al. 2002). Finally, impulsivity is a multifaceted construct that broadly reflects self-regulatory capacity (Gray and MacKillop 2015). It is often cited as a risk factor for the development of AUD, but at the same time, excessive alcohol use can impair judgment and lead to increased impulsivity and risk-taking behavior (Dick et al. 2010). Several measures of impulsivity have been shown to be positively associated with alcohol use including impulsive choice (MacKillop et al. 2007), measured by delayed reward discounting tasks (Kirby et al. 1999), impulsive personality (Magid and Colder 2007; Shin et al. 2012), measured by the Urgency-Premeditation-Perseverance-Sensation Seeking-Positive Urgency (UPPS-P) Impulsive Behavior Scales and Barrett Impulsiveness Scales (Patton et al. 1995), and general personality traits (Cyders et al. 2014). While these mechanisms of risk are well understood to be associated with drinking, the extent to which they are influenced by genetic risk for drinking behavior is not well known.

Current results from large GWASs support the conclusion that alcohol phenotypes are genetically complex and influenced by a large number of genes, suggesting an aggregated measurement of genetic risk can be more powerful than individual variants for association testing. Indeed, polygenic risk scores (PRS) constructed from large meta-GWASs have been robustly associated with alcohol consumption (Ksinan et al. 2019), AUDIT, and its sub-scales (Johnson et al. 2021) in independent samples. Therefore, we chose to include PRSs of drinks per week, PAU, AUD, AUDIT-C, and AUDIT-P to clarify the motivational mechanisms underlying the genetic risk of alcohol use.

Materials and methods

Population Assessment for Tomorrow's Health (PATH) registry

The Population Assessment for Tomorrow's Health (PATH) registry was an initiative at the Peter Boris Centre for Addictions Research to create a database for studies on health behaviors in adults from the greater Hamilton, ON area. Participants were recruited from the general community in Hamilton through flyers and other advertisements ($n = 2164$). They were compensated for their time and received a gift card worth $30 CAD for completing the assessment, as well as a parking voucher or bus tickets to compensate for traveling to the center. Each participant completed psychological measures, underwent a small number of cognitive tests (Supplementary Materials), and provided DNA samples. Eligibility criteria included the following, (i) age 18–65; (ii) $\geq$9th grade education (i.e. adequate literacy to complete written assessments); (iii) willingness to receive invitations for future studies (revocable if preferred); (iv) no current terminal illness (i.e. ability to volunteer to participate in future studies). All participants underwent informed consent, and the registry was approved by the Hamilton Integrated Review Ethics Board (#1074).

Genotyping, imputation, and quality control

Study participants were genotyped on the Illumina Infinium® Global Screening Array by 23andMe Inc using two versions of the array depending on when the saliva samples were received (https://customercare.23andme.com/hc/en-us/articles/218392668-Upgrading-to-23andMe-s-Newest-Chip-Version). Samples from each chip version were imputed separately ($n = 1385$ and 713) using the Michigan Imputation Server and Phase 3 of the 1000 Genomes Project as the reference panel (Das et al. 2016), and merged post-imputation. Only unique SNPs, with INFO score > 0.7, MAF > 0.05, missing rate < 0.01, and Hardy–Weinberg disequilibrium P-value $>1 \times 10^{-5}$ were retained for analysis (5,654,384 SNPs). Genetic ancestry of each sample was determined by the first two principal components using the genotype data combined with those from the 1000 Genomes Project (The 1000 Genomes Project Consortium 2015) as the reference (Supplementary Fig. 1) and compared with self-reported ancestry, only European samples with matching self-report and genetic ancestry were included. We further filtered samples that were within second-degree relationships by retaining at most one member from each pair of samples with a kinship coefficient greater than 0.0884. All analyses were based on a final set of 1534 unrelated, genetically-confirmed European samples, see the consort diagram in Supplementary Fig. 2.

Phenotypic curation

An extensive list of alcohol-related instruments and questions were administered, resulting in 42 distinct high-resolution phenotypes that are categorized into one of the seven phenotypic classes (Table 1). A phenotypic summary of their pairwise correlation patterns can be found in Supplementary Fig. 3.

Alcohol involvement

The AUDIT (SAUNDERS et al. 1993) consists of 10 items, measuring both alcohol consumption frequency (AUDIT-C, items 1 to 3) and problematic use (AUDIT-P, items 4 to 10) within the past year. The Daily Drinking Questionnaire (DDQ) captures consumption of standard drinks within the past year on each day of the week (Collins et al. 1985), rather than asking them to capture consumed drinks on each calendar day using the Timeline Followback Method. Specifically, participants are asked to recall a typical week in the past year and to recall how much they drank a standard drink for each day of the week, providing an average drinking level and typical pattern of drinking for the past year to reduce recall bias. This was then used to calculate the typical number of drinks consumed per week, which was further adjusted for body mass index (BMI); the number of heavy drinking days (HDD) per week using the Canadian Low-Risk Alcohol Drinking Guideline (CLRADG; Butt et al. 2011) defined as 3+/4+ drinks for females/males in one day, respectively; and the proportion of drinks consumed per week according to the CLRADG, whereby 10 drinks for women (15 for men) are the maximum weekly limits (Butt et al. 2011).

Table 1. A description of the 7 phenotypic categories and 42 curated phenotypes.

Category	Measures	Number of phenotypes	Number of phenotypes not imputed for non-users	# of Samples excluded
Alcohol Use	**Historic Alcohol Use:** Age of First Drink	1	0	18
	Consumption (Quantity): AUDIT consumption (first 3 items);	1	0	1
	DDQ HDD per week (CLRADG 'Most Days' definition);	1	0	1
	DDQ Proportion of Drinks per Week (CLRADG definition);	1	0	1
	DDQ Drinks per Week (BMI-adjusted)	1	0	1
	Alcohol-Related Consequences (Severity): AUDIT severity (items 4 to 10)	1	0	1
Alcohol Reinforcing Value	**Proportionate Alcohol-Related Reinforcement:** ALQ;	1	1	305
	Alcohol Demand (Reinforcing Value of Alcohol): APT Indicators (Breakpoint, Elasticity, Intensity, O_{max})	4	1	35 (153 for elasticity)
Reasons for Drinking (Motives)	**DMQ Emotional (Internal) Motivators:** Coping, Enhancement;	2	2	113
	DMQ Socially-Driven (External) Motivators: Conformity, Social	2	2	113
Comorbid Substance Use / Behavioral Disorders	CUDIT total	1	0	1
	DUDIT total	1	0	1
	Cigarette consumption	1	0	1
	Food Addiction (YFAS Symptom Count)	1	0	0
Comorbid Psychopathology	Depression (PHQ-9);	1	0	2
	PTSD (PCL-5);	1	0	0
	WHO ADHD	1	0	0
	Anxiety (PHQ)	1	0	0
	Child Adverse Experience	1	0	0
Impulsivity	**Personality:** UPPS; BIS	11	0	0
	Choice: MCQ (Small, Medium, and Large Magnitudes)	3	0	1, 3, 6 (for small, medium, and large)
Personality	IPIP-NEO (5-factor Personality)	5	0	0

Alcohol reinforcer pathology

A reinforcer pathology approach to PAU emphasizes the high alcohol reinforcing value and the disproportionate reinforcement from drinking in comparison with alternative reinforcers. The former was measured using 4 alcohol demand (i.e. reinforcing value) indices that were quantified using the Alcohol Purchase Task (APT; Minhas et al. 2021): intensity (unrestricted consumption, i.e. zero cost); O_{max} (maximum alcohol expenditure); elasticity (sensitivity to rising cost); and breakpoint (first price when zero-consumption is reached). Processing of the APT indicators is detailed in Supplementary Materials. Alcohol versus alternative reinforcement was measured using the Alcohol Reinforcement Survey Schedule (ARSS; Murphy et al. 2005; Murphy et al. 2007) to calculate the reinforcement ratio of alcohol relative to alcohol-free activities within the past 30 days. A higher reinforcement ratio indicates a higher proportion of psychosocial reinforcement coming from alcohol compared with non-alcohol alternative activities.

Drinking motives

The Drinking Motives Questionnaire - Revised (Cooper 1994) (DMQ-R) consists of 20-items, generating four drinking motives: enhancement, social, conformity, and coping. Enhancement and social motivators are positive reinforcing motives whereas coping and conformity act as negative reinforcing motives (Cooper 1994).

Other addictive behaviors

The Cannabis Use Disorders Identification Test (Adamson et al. 2010) (CUDIT) consists of eight items and provides an overall score representing problematic cannabis use. The Drug Use Disorders Identification Test (Berman et al. 2007) (DUDIT) is made up of 11 items which represent problematic use of illicit and/or pharmaceutical substances, excluding alcohol, cannabis, and tobacco products. Frequency of cigarette use was measured by a single question asking participants to choose the best description of their use over the past year with options 'none', 'monthly', 'weekly', 'daily', and 'multiple times daily'. Compulsive eating was assessed using the Yale Food Addiction Scale 2.0 (Gearhardt et al. 2016) (YFAS); and was included because of its association with PAU and impulsivity (Minhas et al. 2021).

Comorbid psychopathology

Measures of depression, PTSD, and ADHD were used to capture comorbid psychopathology. Both depression and anxiety were assessed using the Patient Health Questionnaire (Kroenke et al. 2001), the former was referred to as the PHQ-9 and the latter was captured using a sum score of three questions from the PHQ that ask about the frequency of feeling nervous, anxious, or on edge. PTSD was assessed using the PTSD Checklist for DSM-5 (Weathers et al. 2013) (PCL-5). ADHD was assessed using the World Health Organization Adult ADHD Self-Report Scale (Kessler et al. 2005). Traumatic childhood experiences were assessed using the ACE questionnaire, which screens for the number of long-term traumatic events the individual was exposed to during their first 18 years of life (Felitti et al. 1998).

Impulsivity

Self-regulatory capacity was captured by measures of impulsive choice or delay discounting, and impulsive personality traits. The former was assessed using the small, medium, and large discounting rates (*k*-values) from the 36-item Extended Monetary Choice Questionnaire (Kirby et al. 1999) (MCQ). The latter was assessed with the Barratt Impulsiveness Scale (Patton et al. 1995) (BIS), a 30-item questionnaire yielding six facets of impulsive personality (attention, cognitive instability, motor, perseverance, self-control, and cognitive complexity) and the S-UPPS-P (Cyders et al. 2014), a 20-item measure with five facets of impulsive personality (negative urgency, positive urgency, sensation seeking, lack of premeditation, and lack of perseverance).

Five-factor personality structure

A 25-item (5-item per personality factor) International Personality Item Pool Representation of the NEO PI-R (Goldberg et al. 2006) (IPIP-NEO) was administered to participants to obtain scores on five key personality factors: openness, conscientiousness, extraversion, agreeableness, and neuroticism.

For the majority of the phenotypes (35/42), individuals that did not endorse drinking were included. For alcohol-related phenotypes, a value of 0 was used for those who did not endorse drinking, such as measures derived from the DDQ and APT. However, $n = 112$ individuals that did not endorse drinking were excluded and treated as missing for (i) alcohol reinforcement ratio as this measure requires the reinforcement of partaking in alcohol-related activities relative to alcohol-free activities; (ii) drinking motivations from the DMQ; and (iii) elasticity from the APT, as this measure is derived from fitting a demand curve, which cannot be calculated for those who have no consumption.

Alcohol SNPs analysis

At the variant level, we examined reported genetic variants with strong evidence of association with any alcohol phenotypes to gain a more comprehensive understanding of the pattern in which risk variants are linked to the high-resolution phenotypic domains. As common genetic variants (with minor allele frequency; MAF > 0.05) tend to have small effect sizes (Park et al. 2011) and considering both the number of variants and sample size of the current study, we considered the variant-level analyses to be complementary and results were reported in the Supplementary Materials Supplementary Tables 6 for completeness.

PRSs association

We obtained GWAS summary statistics for AUDIT-C, and AUDIT-P (Sanchez-Roige et al. 2019c) from the Psychiatric Genomics Consortium (PGC) data portal, drinks per week (Saunders et al. 2022) from the GWAS & Sequencing Consortium of Alcohol and Nicotine use (GSCAN) data portal, PAU (Zhou et al. 2020) and AUD (Kranzler et al. 2019) from the database of Genotypes and Phenotypes (dbGaP; data accession phs001672.v4.p1). Details of these summary statistics can be found in Supplementary Table 2. Quality control steps were performed to match and ensure consistency in the reference alleles between the discovery data and our target. Lassosum (Mak et al. 2017) was used to construct the PRS for its robustness to model misspecification. Construction of the PRSs using Lassosum is detailed in Supplementary Materials. Briefly, we validated each PRS using the alcohol phenotype in PATH that matched most closely to that in the original GWAS. For the AUDIT-C, AUDIT-P, and drinks per week PRSs, the within-sample validation was performed using the exact phenotype. We used a threshold of 8+ on AUDIT total score to match with AUD and the AUDIT-P subscore for PAU. The derived PRSs were then used to test for association with the phenotypes using the following model:

$$z^* \sim PRS + \text{gender} + \text{age} + \text{array} + \text{PC1} + \cdots + \text{PC10}, \quad (1)$$

where z^* was the residual after adjusting for gender, age, the chip version on which the sample was genotyped, and the first 10 genetic principal components, and then transformed using a rank-based transformation to be standard normal. This double adjustment approach was shown to produce unbiased results when the mean of the observed trait is linear in the covariates, but the residual is non-normal (Sofer et al. 2019). For each phenotype, we reported the adjusted R^2 due to PRS as the difference between the full model (PRS and all covariates) and the reduced model (only covariates). We reported a false-discovery rate (FDR) adjusted *P*-value at an α-level of 0.05 for each of the 42 phenotypes examined. All statistical analyses were conducted in Statistical Software R version 4.0.5.

Genetic basis of non-alcohol use phenotypes

Should the association between any alcohol PRSs and the non-alcohol phenotypes in the remaining six domains be significant, we modified the lassosum validation strategy to improve the prediction of the non-alcohol phenotype. Essentially, instead of performing the validation step using an alcohol phenotype that matched most closely to that in the original GWAS, we replaced it with a non-alcohol phenotype. This would generate a list of SNPs from the original alcohol GWAS that had non-zero weights that optimized the prediction of the non-alcohol phenotype, pinpointing genes that overlap in their genetic contribution to phenotypes. The gene mapping was done using FUMA ver. 1.3.8 (Watanabe et al. 2017), returning only genes within 10Kb from the lead SNPs ($P < 5 \times 10^{-8}$) in the original GWASs that also had non-zero weights with respect to each non-alcohol phenotype.

Results

PATH sample demographics and characteristics

The basic demographic characteristics of all enrolled samples and the genetic subsamples are reported in Supplementary Table 1. Across the 42 phenotypes, there was minimal missingness (< 8%) among the 1534 individuals that passed genetic quality controls. The genetic sub-sample was on average older than the overall sample ($P < 0.001$) with a mean age of 34.9 years (SD = 14.2). There was no difference in the proportion of individuals endorsing alcohol consumption between the overall (84.1%) and genetic sub-sample (85.5%), however, the genetic sub-sample on average reported a higher number of typical drinks consumed than the overall sample (8.9+/−12.0 vs. 8.4+/−11.5; $P < 0.001$). As expected, the alcohol use phenotypes were strongly correlated within the domain (Pearson's correlation coefficient $r = 0.43$–0.88), while the correlation was more modest or more variable within drinking motives ($r = 0.27$–0.64) and reinforcing value ($r = 0.21$–0.93) domains (Supplementary Fig. 3). When using AUDIT-C and AUDIT-P as representative alcohol use phenotypes, we found that drinking motives and reinforcing values to be most strongly associated with alcohol use, with the exception of conformity as a motive (Supplementary Fig. 4). Though impulsivity and other comorbid conditions did not show a strong correlation, Positive Urgency and DUDIT were moderately correlated with both AUDIT-C and AUDIT-P, but exhibited a stronger relationship with AUDIT-P.

Summary of PRSs performance

All PRSs, except PRS of AUD, were significantly associated with the target trait after FDR correction. The final lassosum alcohol PRSs, namely, $PRS_{AUDIT\text{-}C}$, $PRS_{AUDIT\text{-}P}$, $PRS_{drinks\text{-}per\text{-}week}$, and PRS_{PAU}, each included 2,338,199, 384,933, 221,689, and 453,914 autosomal SNPs, and explained 1.35, 0.46, 1.68, and 0.35% of the phenotypic variance for the target trait in PATH (Supplementary Table 3). Across the 42 phenotypes, $PRS_{drinks\text{-}per\text{-}week}$ was significantly associated with 15 phenotypes (FDR adjusted P-values <0.05), $PRS_{AUDIT\text{-}C}$ and $PRS_{AUDIT\text{-}P}$ were each associated with 13 and 10 phenotypes, respectively, and PRS_{PAU} was associated with seven phenotypes. Since the PRS of AUD was not significantly associated with within-sample AUD in PATH nor any of the 42 broad phenotypes, we focused on results using $PRS_{AUDIT\text{-}C}$, $PRS_{AUDIT\text{-}P}$, $PRS_{drinks\text{-}per\text{-}week}$, and PRS_{PAU}.

PRSs associations with alcohol-specific domains

All four PRSs were positively associated with phenotypes in the alcohol-specific phenotypic domains (FDR adjusted P-values <0.05), including general alcohol use, alcohol reinforcing value, and drinking motives (Fig. 1 and Table 2). For phenotypes in the alcohol use category, $PRS_{AUDIT\text{-}C}$ and $PRS_{drinks\text{-}per\text{-}week}$ had overall higher adjusted R^2 values of 0.7–1.52 and 0.71–1.68%, respectively, than $PRS_{AUDIT\text{-}P}$ (0.46 to 0.51%) and PRS_{PAU} (0.43 to 0.79%). As expected, $PRS_{AUDIT\text{-}C}$ and $PRS_{drinks\text{-}per\text{-}week}$ were more significantly associated with alcohol involvement and their strongest associations were with proportion of drinks per week and BMI-adjusted daily drinking quantity (FDR adjusted P-values $<3.3 \times 10^{-5}$), explaining 1.36–1.68% of the phenotypic variance. While $PRS_{AUDIT\text{-}P}$ explained a moderate amount of variance across all alcohol use phenotypes, PRS_{PAU} showed an elevated effect with HDD per week, explaining 0.79% of its phenotypic variance.

All PRSs were implicated with at least one phenotype from the drinking motives and alcohol reinforcing value domains. Again, $PRS_{AUDIT\text{-}C}$ and $PRS_{drinks\text{-}per\text{-}week}$ showed similar patterns of positive association with both the negative motive of coping (adjusted R^2s = 0.45 and 0.52%) and the positive motives of social (adjusted R^2s = 0.7 and 1.01%) and enhancement (adjusted R^2s = 0.7 and 2.28%), but differed slightly with $PRS_{drinks\text{-}per\text{-}week}$ exhibiting a much larger effect on phenotypes in the drinking motives category. PRS_{PAU} was only associated with enhancement and $PRS_{AUDIT\text{-}P}$ was additionally associated with the social motive with only moderate effects (adjusted R^2s = 0.41–0.53%).

Across phenotypes in the alcohol reinforcing category, all except breakpoint were associated with both $PRS_{AUDIT\text{-}C}$ and $PRS_{drinks\text{-}per\text{-}week}$. The strength of associations between alcohol reinforcing mechanisms and $PRS_{AUDIT\text{-}C}$, was again dominated by $PRS_{drinks\text{-}per\text{-}week}$ for all (adjusted R^2s = 0.43–0.93% vs. adjusted R^2s = 0.91–1.33%) but alcohol reinforcement ratio (adjusted R^2 = 1.33% vs. 1.29%). Alcohol reinforcement ratio was not associated with $PRS_{AUDIT\text{-}P}$ or PRS_{PAU}, but both were negatively associated with elasticity (adjusted R^2s = 0.43 and 0.74%). $PRS_{AUDIT\text{-}P}$ was also associated with intensity and Omax with moderate effects (adjusted R^2s = 0.36–0.38%).

PRS associations with co-occurring conditions

While $PRS_{AUDIT\text{-}P}$ was not associated with any of the co-occurring condition, $PRS_{AUDIT\text{-}C}$ was associated with cigarette use (adjusted R^2 = 0.46). Both PRS_{PAU} and $PRS_{drinks\text{-}per\text{-}week}$ were positively linked to the ACE score (adjusted R^2s = 0.73 and 0.6%), and $PRS_{drinks\text{-}per\text{-}week}$ was additionally positively associated with WHO ADHD part A score, albeit weakly (adjusted R^2 = 0.34).

PRS associations with impulsivity and personality

There were only two associations identified for impulsivity, linking the attention factor of BIS to $PRS_{drinks\text{-}per\text{-}week}$ and the complexity factor of BIS to PRS_{PAU}, explaining 0.43 and 0.41% of the phenotypic variance, respectively. None of the personality traits were significantly associated with any PRSs.

Genetic basis of non-alcohol use phenotypes

The observed association between alcohol PRSs and motivational mechanisms could be manifested in several ways. We found nine non-alcohol phenotypes in four domains (drinking motives, reinforcing pathology, comorbid conditions, and impulsivity) that showed evidence of association with at least one of the alcohol PRSs. The strength of association between each of the nine phenotypes and their corresponding optimized PRSs improved as compared with that using alcohol PRSs (Supplementary Table 5). Notably, DMQ-social required the least number of non-zero SNPs to improve the prediction with respect to all four alcohol PRSs. On the other hand, in terms of the number of genes implicated, it ranged from 326 for DMQ-social to 404 for Omax (Supplementary Table 6), which was the maximum number of unique genes identified across the original GWASs (Supplementary Fig. 5).

Discussion

The current study investigated the functional significance of genetic signals implicated with alcohol involvement from large-scale GWAS studies in a broader phenotypic context. We showed that alcohol consumption PRSs, namely, $PRS_{drinks\text{-}per\text{-}week}$ and $PRS_{AUDIT\text{-}C}$ were significantly associated with within-sample AUDIT-C score and other drinking indicators (drinks per week, HDD, proportion of drinks per week), reaffirming previous findings that a PRS of AUDIT-C explained between 0.4 and 0.96% of alcohol use or dependence phenotype variance in other studies (Sanchez-Roige et al. 2019a; Johnson et al. 2021). More importantly,

Table 2. Genetic association of AUDIT-C and AUDIT-P PRSs with screened phenotypes.

Category	Phenotype	AUDIT-C PRS			AUDIT-P PRS			GSCAN - drinks per week PRS			PAU PRS		
		Effect	adjR2 (%)	*P*-value	Effect	adjR2 (%)	*P*-value	Effect	adjR2 (%)	*P*-value	Effect	adjR2 (%)	*P*-value
Alcohol Use	Proportion of Drinks per Week	0.13	1.52	8.49E-07	0.07	0.49	0.004	0.12	1.36	3.06E-06	0.07	0.44	0.006
	HDD per week	0.1	1.00	5.54E-05	0.08	0.51	0.003	0.09	0.71	5.92E-04	0.09	0.79	3.07E-04
	Drinks per Week (BMI-adjusted)	0.12	1.47	1.42E-06	0.08	0.51	0.003	0.13	1.68	3.65E-07	0.07	0.43	0.007
	AUDIT-Problem	0.09	0.70	6.70E-04	0.07	0.46	0.005	0.1	0.95	8.44E-05	0.06	0.35	0.01
	AUDIT-Consumption	0.12	1.35	3.27E-06	0.07	0.46	0.005	0.12	1.48	1.17E-06	0.05	0.19	0.05
Drinking Motives	DMQ-Social	0.09	0.70	9.98E-04	0.07	0.44	0.007	0.11	1.06	5.93E-05	0.06	0.27	0.03
	DMQ-Coping	0.07	0.45	0.006	0.04	0.06	0.18	0.08	0.52	0.004	0.04	0.13	0.09
	DMQ-enhancement	0.09	0.71	8.51E-04	0.07	0.41	0.009	0.15	2.28	6.62E-09	0.08	0.53	0.003
	DMQ-conformity	0.03	0.05	0.2	0.04	0.07	0.14	0	−0.07	0.95	0.01	−0.07	0.77
Alcohol Reinforcing Value	Alcohol Reinforcement Ratio	0.12	1.33	3.63E-05	0.05	0.14	0.11	0.12	1.29	4.54E-05	0.03	0	0.32
	Elasticity	−0.07	0.43	0.009	−0.07	0.46	0.007	−0.12	1.33	1.14E-05	−0.09	0.74	9.33E-04
	Breakpoint	0.06	0.30	0.02	0.04	0.07	0.15	0.06	0.29	0.02	0.02	−0.04	0.5
	Intensity	0.1	0.93	1.18E-04	0.07	0.38	0.01	0.11	1.08	3.54E-05	0.05	0.2	0.05
	Omax	0.09	0.68	8.59E-04	0.06	0.36	0.01	0.1	0.91	1.36E-04	0.05	0.23	0.04
Addictive Behaviors	Cigarettes per day	0.07	0.46	0.005	0.06	0.25	0.03	0.05	0.14	0.08	0.06	0.28	0.02
Comorbid Psychiatric Syndromes	WHO ADHD	0.02	−0.02	0.41	0.03	0.02	0.25	0.06	0.34	0.01	0.03	0.04	0.2
	ACE	0.01	−0.06	0.69	0.06	0.26	0.03	0.08	0.6	0.001	0.09	0.73	4.79E-04
Impulsivity	Attention (BIS)	0.04	0.09	0.13	0.03	0.01	0.29	0.07	0.43	0.006	0.02	−0.04	0.54
	Complexity (BIS)	0.03	0.01	0.27	0.04	0.08	0.14	0.05	0.15	0.07	0.07	0.41	0.007

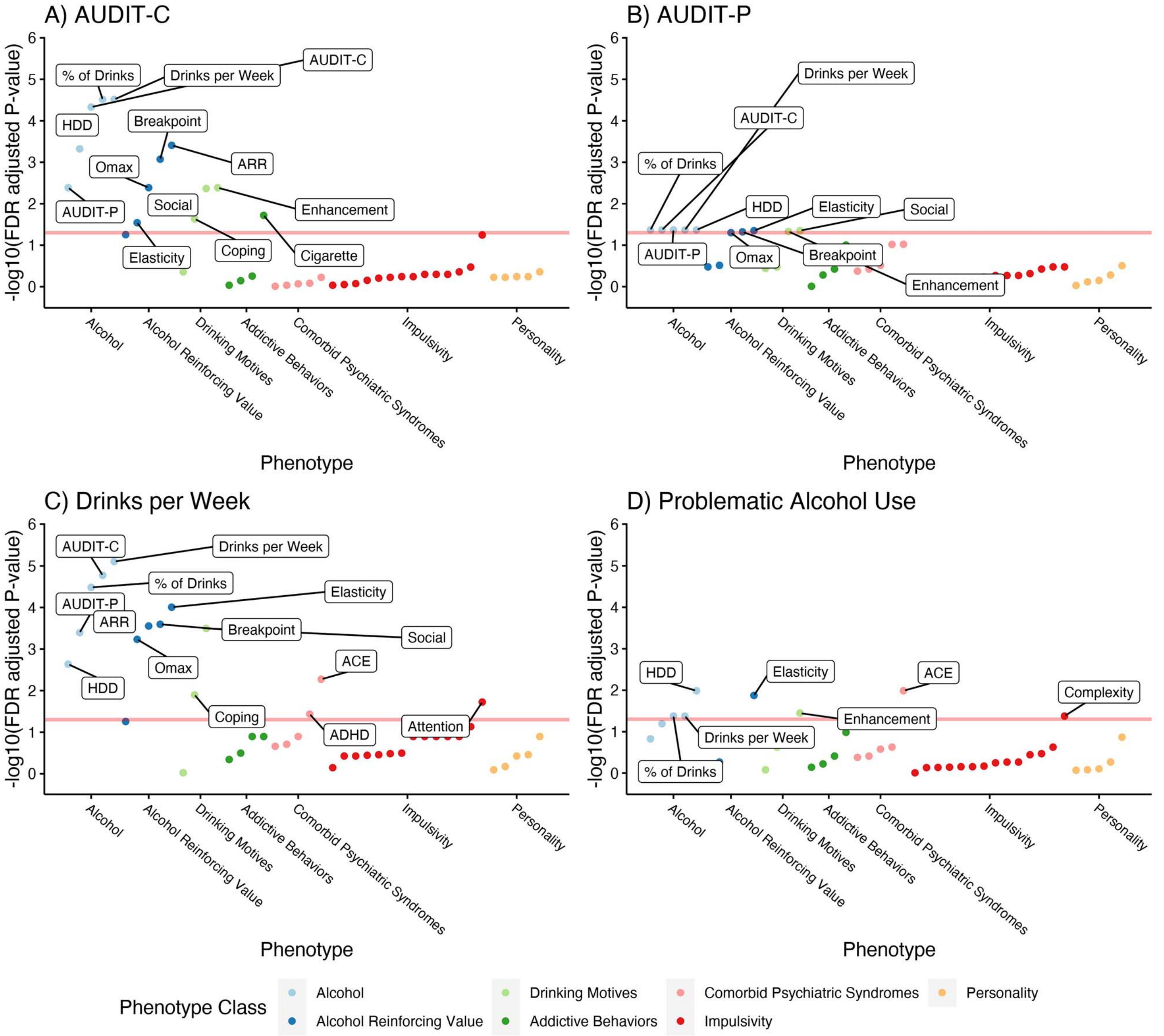

Figure 1 PheWAS results of associations between AUDIT-C, AUDIT-P, drinks per week, PAU, addiction, and AUD PRSs **and 42 curated phenotypes.** The -log10 false discovery rate adjusted *P*-value is plotted against each of the 42 phenotypes in the seven phenotypic classes shown on the *X*-axis. The phenotypic classes on the left of the *X*-axis are alcohol-specific and increasingly less specific as we move toward the right. The horizontal lines indicate the threshold at a false discovery rate adjusted *P*-value < 0.05.

$PRS_{AUDIT\text{-}C}$ and $PRS_{drinks\text{-}per\text{-}week}$ were linked for the first time to alcohol reinforcing value (elasticity, demand intensity, Omax, and reinforcement ratio), positive drinking motives of social and enhancement, and also the negative drinking motive of coping. These suggest that individuals with higher genetic predisposition to alcohol involvement find alcohol more reinforcing (i.e. having greater incentive salience), presumably due to biological differences in either alcohol pharmacokinetics or pharmacodynamics. Such differences could result in either more positively or negatively reinforcing pharmacological properties (i.e. rewarding or relieving effects) or reduced unpleasant consequences resulting from metabolic by-products, such as acetaldehyde. Higher reinforcing value could translate to a higher risk of excessive consumption and potentially the development of AUD and, consistent with this, problematic use PRSs, namely, $PRS_{AUDIT\text{-}P}$ and PRS_{PAU}, had on average weaker associations with quantity of alcohol use, but showed specificity in their associations with elasticity, intensity, and Omax of the APT, which are associated with heavy alcohol use and alcohol-related problems (Martínez-Loredo et al. 2021). In contrast, social motives were strongly correlated with alcohol consumption PRSs, but not with PRS_{PAU}, which reveals a genetic signal for social drinking, but not alcohol problems. These results highlight a varied set of motivational mechanisms captured by the genetics of alcohol involvement.

Remarkably, both $PRS_{drinks\text{-}per\text{-}week}$ and PRS_{PAU} were associated with the ACE score that also partially reflects family history of alcohol misuse. Indeed, an item-by-item analysis revealed that the association was strongest with the following item: ‘Did you live with anyone who was a problem drinker or alcoholic or who used street drugs?’, where $PRS_{drinks\text{-}per\text{-}week}$ explained 0.88% ($P = 0.0001$) of the variance vs. 0.6% of the total score variance ($P = 0.0015$). Meanwhile, the strongest association with PRS_{PAU} came from items 5 ($P = 0.0016$) and 7 ($P = 0.00051$), where item 5 captures neglect related to alcohol misuse (‘Your parents were too drunk or high to take care of you or take you to the doctor if you needed it?') and item 7 captures a history of physical violence from maternal caregivers. These results are interesting in that the genetic correlation between ACE scores and alcohol use in later adulthood could potentially have been overestimated due to both the family history of alcohol misuse as well as the adverse experiences resulting from it.

Unlike $PRS_{drinks\text{-}per\text{-}week}$ and PRS_{PAU}, the two-subscales of AUDIT, particularly AUDIT-P, did not map onto almost any non-alcohol specific domains possibly because these non-alcohol specific factors are weakly correlated with alcohol use phenotypically (Supplementary Fig. 4). While the AUDIT total scores have been reported to have weak to moderate genetic correlation with comorbid psychiatric syndromes (Sanchez-Roige et al. 2019c) and impulsive personality traits (Sanchez-Roige et al. 2019b; 2023), the lack of observed PRS associations could be a power issue. Another possible explanation could be that AUDIT-P, calculated as a sum of scores from seven items, is an inherently more heterogenous phenotype than AUDIT-C and requires carefully tuned weightings of these itemized scores to produce an improved polygenic signal (Mallard and Sanchez-Roige 2021). Indeed, similar conclusions on the different performance of PRSs for the two subscales have been reached using other methods and in other samples (Sanchez-Roige et al. 2019a; Johnson et al. 2021), collectively pointing to the complex sources of genetic influence on AUDIT-C and AUDIT-P contributing to AUD/PAU and the need to better categorize alcohol phenotypes. For example, though Sensation Seeking and Positive Urgency are genetically linked to alcohol consumption (Sanchez-Roige et al. 2019a), they do not correlate with DSM-IV diagnosis of alcohol dependence (Walters et al. 2018), supporting a categorization of alcohol phenotypes through identifying shared and differential genetic signals from non-alcohol specific mechanisms.

These results should be interpreted with the following considerations in mind. The current study ($n = 1534$) was not sufficiently powered to map certain alcohol PRS onto all non-alcohol specific domains at the FDR corrected significance level, even though there was consistent suggestive evidence across both phenotypic domains and PRSs. Specifically, Cigarettes per day showed consistently effects across all alcohol PRSs with P-values between 0.08 and 0.005, but only was significantly associated with AUDIT-C. This suggests limited power, but another plausible explanation, perhaps more consistent with, is the presence of regional genetic correlation involving certain genes or specific genomic regions, for example the dopamine receptor D2 (*DRD2*), *the brain-derived neurotrophic factor* (*BDNF*) gene, and the cell adhesion molecule 2 (*CADM2*) gene region (Saunders et al. 2022). In addition, the associated effects were on average stronger for positive reinforcement phenotypes than negative reinforcement phenotypes and comorbid conditions; these could potentially reflect bias from the study sample in a general population that was not enriched for adults with addiction and/or mental health issues. Finally, to reduce the confounding effect of population stratification, this study included only genetically European participants and thus it is uncertain how findings will generalize to other ancestry groups.

In sum, in contextualizing the phenotypic influence of genetic signals from large-scale GWAS studies of alcohol use, the current investigation found notable links with positive reinforcement phenotypes (alcohol demand, enhancement, and social motives), negative reinforcement phenotypes (e.g. coping motives, comorbidities), and impulsivity-related phenotypes. These findings offer new clues into the risk mechanisms by which genetic variants that collectively associated with alcohol may confer risk for alcohol misuse, indicating a multiplicity of domains other than alcohol itself. Verification of these links through further application of a high-resolution PheWAS in larger and more diverse samples is warranted.

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Authors' contributions

W.Q.D. and J.M. were responsible for the study concept and design. W.Q.D. performed the genomic analysis. K.B. performed phenotypic analysis. All authors drafted the manuscript, critically reviewed the content, provided revisions of the manuscript for important intellectual content, and approved the final version.

Supplementary material

Supplementary material is available at *Alcohol and Alcoholism* online.

Conflict of interest: J.M. is a principal and senior scientist in Beam Diagnostics, Inc and has consulted to Clairvoyant Therapeutics, Inc.

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Data availability

The summary statistics for the two AUDIT subscales are available from the PGC data portal (https://www.med.unc.edu/pgc/download-results/). The drinks per week summary statistics were obtained from the GWAS & Sequencing Consortium of Alcohol and Nicotine use (GSCAN) data portal (https://conservancy.umn.edu/handle/11299/241912). Summary statistics for problematic alcohol use and alcohol use disorder were derived partially using data from Million Veteran Program and were available from the database of Genotypes and Phenotypes (dbGaP; data accession phs001672.v4.p1). The full results of SNP analysis and polygenic risk scores derived in the current manuscript will be made available at https://github.com/PBCAR.

Disclaimer

The opinions and assertions expressed herein are those of the authors and do not necessarily reflect the official policy or position of the Uniformed Services University or the Department of Defense.

References

Adamson SJ, Kay-Lambkin FJ, Baker AL. *et al.* An improved brief measure of cannabis misuse: the Cannabis Use Disorders Identification Test-Revised (CUDIT-R). *Drug Alcohol Depend* 2010;**110**:137–43. https://doi.org/10.1016/j.drugalcdep.2010.02.017.

Berman AH, Palmstierna T, Källmén H. *et al.* The self-report Drug Use Disorders Identification Test-Extended (DUDIT-E): reliability, validity, and motivational index. *J Subst Abuse Treat* 2007;**32**: 357–69. https://doi.org/10.1016/j.jsat.2006.10.001.

Blanco C, Xu Y, Brady K. *et al.* Comorbidity of posttraumatic stress disorder with alcohol dependence among US adults: results from national epidemiological survey on alcohol and related conditions. *Drug Alcohol Depend* 2013;**132**:630–8. https://doi.org/10.1016/j.drugalcdep.2013.04.016.

Butt P, Beirness D, Gliksman L. *et al. Alcohol and health in Canada: a summary of evidence and guidelines for low-risk drinking*. Canadian Centre on Substance Abuse, 2011.

Cavicchioli M, Vassena G, Movalli M. *et al.* Addictive behaviors in alcohol use disorder: dysregulation of reward processing systems and maladaptive coping strategies. *J Addict Dis* 2018;**37**:173–84. https://doi.org/10.1080/10550887.2019.1643211.

Céspedes IC, Ota VK, Mazzotti DR. *et al.* Association between polymorphism in gene related to the dopamine circuit and motivations for drinking in patients with alcohol use disorder. *Psychiatry Res* 2021;**295**:113563. https://doi.org/10.1016/j.psychres.2020.113563.

Clarke TK, Adams MJ, Davies G. *et al.* Genome-wide association study of alcohol consumption and genetic overlap with other health-related traits in UK biobank (N=112117). *Mol Psychiatry* 2017;**22**: 1376–84. https://doi.org/10.1038/mp.2017.153.

Collins RL, Parks GA, Marlatt GA. Social determinants of alcohol consumption. The effects of social interaction and model status on the self-administration of alcohol. *J Consult Clin Psychol* 1985;**53**(2):189–200. https://doi.org/10.1037/0022-006X.53.2.189.

Cooper ML. Motivations for alcohol use among adolescents: development and validation of a four-factor model. *Psychol Assess* 1994;**6**(2):117–128. https://doi.org/10.1037/1040-3590.6.2.117.

Cyders MA, Littlefield AK, Coffey S. *et al.* Examination of a short English version of the UPPS-P Impulsive Behavior Scale. *Addict Behav* 2014;**39**:1372–6. https://doi.org/10.1016/j.addbeh.2014.02.013.

Das S, Forer L, Schönherr S. *et al.* Next-generation genotype imputation service and methods. *Nat Genet* 2016;**48**:1284–7. https://doi.org/10.1038/ng.3656.

Debell F, Fear NT, Head M. *et al.* A systematic review of the comorbidity between PTSD and alcohol misuse. *Soc Psychiatry Psychiatr Epidemiol* 2014;**49**:1401–25. https://doi.org/10.1007/s00127-014-0855-7.

Dick DM, Smith G, Olausson P. *et al.* Understanding the construct of impulsivity and its relationship to alcohol use disorders. *Addict Biol* 2010;**15**:217–26. https://doi.org/10.1111/j.1369-1600.2009.00190.x.

Dube SR, Anda RF, Felitti VJ. *et al.* Adverse childhood experiences and personal alcohol abuse as an adult. *Addict Behav* 2002;**27**(5):713–725. https://doi.org/10.1016/S0306-4603(01)00204-0.

Felitti VJ, Anda RF, Nordenberg D. *et al.* Relationship of childhood abuse and household dysfunction to many of the leading causes of death in adults: the adverse childhood experiences (ACE) study. *Am J Prev Med* 1998;**14**(4):245–258. https://doi.org/10.1016/S0749-3797(98)00017-8.

Gearhardt AN, Corbin WR, Brownell KD. Development of the Yale Food Addiction Scale Version 2.0. *Psychol Addict Behav* 2016;**30**(1):113–121. https://doi.org/10.1037/adb0000136.

Gelernter J, Kranzler HR, Sherva R. *et al.* Genome-wide association study of alcohol dependence: significant findings in African- and European-Americans including novel risk loci. *Mol Psychiatry* 2014;**19**:41–9. https://doi.org/10.1038/mp.2013.145.

Goldberg LR, Johnson JA, Eber HW. *et al.* The international personality item pool and the future of public-domain personality measures. *J Res Pers* 2006;**40**(1):358. https://doi.org/10.1016/j.jrp.2005.08.007.

Gray JC, MacKillop J. Using behavior economics to understand alcohol use disorders: a concise review and identification of research priorities. *Curr Addict Rep* 2015;**2**:68–75. https://doi.org/10.1007/s40429-015-0045-z.

Johnson EC, Sanchez-Roige S, Acion L. *et al.* Polygenic contributions to alcohol use and alcohol use disorders across population-based and clinically ascertained samples. *Psychol Med* 2021;**51**:1147–56. https://doi.org/10.1017/S0033291719004045.

Kathryn, Mchugh R, Weiss RD. Alcohol use disorder and depressive disorders. *Alcohol Res* 2019;**40**(1):arcr.v40.1.01. https://doi.org/10.35946/arcr.v40.1.01.

Kessler RC, Adler L, Ames M. *et al.* The World Health Organization adult ADHD self-report scale (ASRS): a short screening scale for use in the general population. *Psychol Med* 2005;**35**(2):245–256. https://doi.org/10.1017/S0033291704002892.

Kirby KN, Petry NM, Bickel WK. Heroin addicts have higher discount rates for delayed rewards than non-drug-using controls. *J Exp Psychol Gen* 1999;**128**(1):78–87. https://doi.org/10.1037/0096-3445.128.1.78.

Kranzler HR, Zhou H, Kember RL. *et al.* Genome-wide association study of alcohol consumption and use disorder in 274,424 individuals from multiple populations. *Nat Commun* 2019;**10**(1):1499. https://doi.org/10.1038/s41467-019-09480-8.

Kroenke K, Spitzer RL, Williams JBW. The PHQ-9: validity of a brief depression severity measure. *J Gen Intern Med* 2001;**16**:606–13. https://doi.org/10.1046/j.1525-1497.2001.016009606.x.

Ksinan AJ, Su J, Aliev F; Spit for Science Workgroup, Dick DM. Unpacking Genetic Risk Pathways for College Student Alcohol Consumption: The Mediating Role of Impulsivity. *Alcohol Clin Exp Res*. 2019;**43**(10):2100–2110. https://doi.org/10.1111/acer.14157.

Liu M, Jiang Y, Wedow R. *et al.* Association studies of up to 1.2 million individuals yield new insights into the genetic etiology of tobacco and alcohol use. *Nat Genet* 2019;**51**:237–44. https://doi.org/10.1038/s41588-018-0307-5.

MacKillop J, Mattson RE, MacKillop EJA. *et al.* Multidimensional assessment of impulsivity in undergraduate hazardous drinkers and controls. *J Stud Alcohol Drugs* 2007;**68**(6):785–788. https://doi.org/10.15288/jsad.2007.68.785.

MacKillop J, Agabio R, Feldstein Ewing SW. *et al.* Hazardous drinking and alcohol use disorders. *Nat Rev Dis Primers* 2022;**8**(1):80. https://doi.org/10.1038/s41572-022-00406-1.

Magid V, Colder CR. The UPPS Impulsive Behavior Scale: factor structure and associations with college drinking. *Pers Individ Dif* 2007;**43**(7): 1927–1937. https://doi.org/10.1016/j.paid.2007.06.013.

Mak TSH, Porsch RM, Choi SW. *et al.* Polygenic scores via penalized regression on summary statistics. *Genet Epidemiol* 2017;**41**: 469–80. https://doi.org/10.1002/gepi.22050.

Mallard TT, Sanchez-Roige S. Dimensional phenotypes in psychiatric genetics: lessons from genome-wide association studies of alcohol use phenotypes. *Complex Psychiatry* 2021;**7**:45–8. https://doi.org/10.1159/000518863.

Mallard TT, Savage JE, Johnson EC. *et al.* Item-level genome-wide association study of the alcohol use disorders identification test in three population-based cohorts. *Am J Psychiatry* 2022;**179**:58–70. https://doi.org/10.1176/appi.ajp.2020.20091390.

Martínez-Loredo V, González-Roz A, Secades-Villa R. *et al.* Concurrent validity of the Alcohol Purchase Task for measuring the reinforcing efficacy of alcohol: an updated systematic review and meta-analysis. *Addiction* 2021;**116**:2635–50. https://doi.org/10.1111/add.15379.

Minhas M, Murphy CM, Balodis IM. *et al.* Multidimensional elements of impulsivity as shared and unique risk factors for food addiction and alcohol misuse. *Appetite* 2021;**159**:105052. https://doi.org/10.1016/j.appet.2020.105052.

Murphy JG, MacKillop J. Relative reinforcing efficacy of alcohol among college student drinkers. *Exp Clin Psychopharmacol* 2006;**14**: 219–27. https://doi.org/10.1037/1064-1297.14.2.219.

Murphy JG, Colby SM, Correia CJ. *et al.* Using behavioral theories of choice to predict drinking outcomes following a brief intervention. *Exp Clin Psychopharmacol* 2005;**13**:93–101. https://doi.org/10.1037/1064-1297.13.2.93.

Murphy JG, Correia CJ, Barnett NP. Behavioral economic approaches to reduce college student drinking. *Addict Behav* 2007;**32**(11):2573–2585. https://doi.org/10.1016/j.addbeh.2007.05.015.

Ohlmeier MD, Peters K, te Wildt BT. *et al.* Comorbidity of alcohol and substance dependence with attention-deficit/ hyperactivity disorder (ADHD). *Alcohol Alcohol* 2008;**43**(3):300–304. https://doi.org/10.1093/alcalc/agn014.

Park JH, Gail MH, Weinberg CR. *et al.* Distribution of allele frequencies and effect sizes and their interrelationships for common genetic susceptibility variants. *Proc Natl Acad Sci U S A* 2011;**108**:18026–31. https://doi.org/10.1073/pnas.1114759108.

Patton JH, Stanford MS, Barratt ES. Factor structure of the Barratt impulsiveness scale. *J Clin Psychol* 1995;**51**(6):768–774. https://doi.org/10.1002/1097-4679(199511)51:6<768::AID-JCLP2270510607>3.0.CO;2-1.

Sanchez-Roige S, Palmer AA, Fontanillas P. *et al.* Genome-wide association study meta-analysis of the alcohol use disorders identification test (AUDIT) in two population-based cohorts. *Am J Psychiatry* 2019c;**176**:107–18. https://doi.org/10.1176/appi.ajp.2018.18040369.

Sanchez-Roige S, Fontanillas P, Elson SL. *et al.* Genome-wide association study of alcohol use disorder identification test (AUDIT) scores in 20 328 research participants of European ancestry. *Addict Biol* 2019a;**24**:121–31. https://doi.org/10.1111/adb.12574.

Sanchez-Roige S, Fontanillas P, Elson SL. *et al.* Genome-wide association studies of impulsive personality traits (BIS-11 and UPPS-P) and drug experimentation in up to 22,861 adult research participants identify loci in the CACNA1I and CADM2 genes. *J Neurosci* 2019b;**39**:2562–72. https://doi.org/10.1523/JNEUROSCI.2662-18.2019.

Sanchez-Roige S, Jennings MV, Thorpe HHA. *et al.* CADM2 is implicated in impulsive personality and numerous other traits by genome- and phenome-wide association studies in humans and mice. *Transl Psychiatry* 2023; **13**:167. https://doi.org/10.1038/s41398-023-02453-y.

Saunders JB, Aasland OG, Babor TF. *et al.* Development of the Alcohol Use Disorders Identification Test (AUDIT): WHO collaborative project on early detection of persons with harmful alcohol consumption-II. *Addiction* 1993;**88**:791–804. https://doi.org/10.1111/j.1360-0443.1993.tb02093.x.

Saunders GRB, Wang X, Chen F. *et al.* Genetic diversity fuels gene discovery for tobacco and alcohol use. *Nature* 2022;**612**:720–4. https://doi.org/10.1038/s41586-022-05477-4.

Shin SH, Hong HG, Jeon SM. Personality and alcohol use: the role of impulsivity. *Addict Behav* 2012;**37**:102–7. https://doi.org/10.1016/j.addbeh.2011.09.006.

Sofer T, Zheng X, Gogarten SM. *et al.* A fully adjusted two-stage procedure for rank-normalization in genetic association studies. *Genet Epidemiol* 2019;**43**:263–75. https://doi.org/10.1002/gepi.22188.

The 1000 Genomes Project Consortium. A global reference for human genetic variation. The 1000 Genomes Project Consortium∗. *Nature* 2015;**526**:68–74. https://doi.org/10.1038/nature15393.

Tsuang MT, Bar JL, Harley RM. *et al.* The Harvard Twin Study of Substance Abuse: what we have learned. *Harv Rev Psychiatry* 2001;**9**(6):267–279. https://doi.org/10.1080/10673220127912.

Walters RK, Polimanti R, Johnson EC. *et al.* Transancestral GWAS of alcohol dependence reveals common genetic underpinnings with psychiatric disorders. *Nat Neurosci* 2018;**21**:1656–69. https://doi.org/10.1038/s41593-018-0275-1.

Watanabe K, Taskesen E, Van Bochoven A. *et al.* Functional mapping and annotation of genetic associations with FUMA. *Nat Commun* 2017;**8**:1826. https://doi.org/10.1038/s41467-017-01261-5.

Weathers FW, Blake DD, Schnurr PP. *et al. The Life Events Checklist for DSM-5 (LEC-5)*. 2013, Instrument available from the National Center for PTSD at www.ptsd.va.gov.

Zhou H, Sealock JM, Sanchez-Roige S. *et al.* Genome-wide meta-analysis of problematic alcohol use in 435,563 individuals yields insights into biology and relationships with other traits. *Nat Neurosci* 2020;**23**:809–18. https://doi.org/10.1038/s41593-020-0643-5.